

INCIDENT INVESTIGATION REPORT: ASSET P-194

1. Incident Summary

Asset Tag: P-194 (Centrifugal Heavy-Duty Process Pump)
Incident Date: 2026-02-14
Failure Mode: Bearing Thermal Distress & High Vibration Trip
Unplanned Downtime: 4.2 hours
Assigned Work Order: WO-9412

2. Sequence of Events & Root Cause Analysis

At 14:15, vibration telemetry on P-194 spiked to 9.8 mm/s RMS (trip setpoint 7.1 mm/s).
Drive-end bearing temperature reached 92.4 deg C.
Immediate emergency shutdown prevented catastrophic impeller seizure.
Root Cause: Oil seal degradation permitted boundary weepage and progressive lubricant starvation.
Metal-to-metal contact generated localized micro-spalling on the inner raceway.

3. Corrective Actions & Resolution

Work Order WO-9412 executed emergency bearing replacement.
Fitted new SKF-6314-2Z drive-end bearing and renewed BURG-M7N mechanical seal.
Laser alignment verified to within 0.03 mm precision tolerance.
Pump recommissioned successfully with baseline vibration at 2.1 mm/s RMS.